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# A Comparative Analysis on Feature Selection for Regression Models: Comparing SHAP-Based Methods with Traditional Approaches

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# Abstract

This thesis investigates whether SHAP-based feature selection offers practical advantages over traditional methods for tabular regression. A two-stage experimental protocol compares SHAP importance against mutual information, recursive feature elimination, and tree-based importance across multiple datasets and model families.

The findings indicate that SHAP-based selection is competitive but not systematically superior, while incurring additional computational cost. Hybrid strategies combining native importance with SHAP refinement do not yield meaningful improvements over simpler approaches.

These results suggest that SHAP remains most valuable as an explanatory tool rather than a selection criterion, and that practitioners should prefer simpler methods unless task-specific evidence justifies the overhead.

**Keywords:** feature selection, SHAP, regression, machine learning, model interpretability, TreeExplainer.

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# Chapter 1

# Introduction

Machine learning has transformed how organizations extract value from data, enabling predictive models that automate decisions in domains ranging from financial risk assessment to industrial process optimization. Among the many tasks addressed by machine learning, regression (the prediction of continuous outcomes) remains fundamental. Whether estimating insurance claim costs, forecasting energy demand, or predicting material properties, regression models serve as the backbone of quantitative decision support systems.

As datasets grow in both volume and dimensionality, practitioners face an increasingly common challenge: determining which input variables actually contribute to accurate predictions. High-dimensional datasets may contain hundreds or thousands of features, yet only a subset typically carries meaningful predictive signal. The remaining features introduce noise, increase computational burden, and complicate model interpretation. Feature selection addresses this challenge by identifying and retaining only the most informative variables before or during model training.

## 1.1 Context and Motivation

The importance of feature selection extends beyond dimensionality reduction. Effective selection can improve generalization by reducing overfitting, decrease training and inference time, lower storage, and when applied with model explanation strategies, produce models that are easier to explain and maintain. In regulated industries or high-stakes applications, the ability to articulate why a model uses specific inputs is often as important as the accuracy of its predictions.

Traditional feature selection methods fall into three categories: filter methods that rank features by statistical criteria independently of the model, wrapper methods that evaluate subsets through iterative model training, and embedded methods where selection occurs as part of the learning algorithm itself [8]. Each approach involves trade-offs between computational cost, model dependency, and theoretical guarantees.

Model interpretability tools have gained prominence, with SHAP (SHapley Additive exPlanations) emerging as a influential framework [15]. SHAP provides theoretically grounded feature attributions derived from cooperative game theory [22], offering both local explanations for individual predictions and global importance rankings across datasets [5]. While SHAP is primarily designed for explanation, its global importance scores can be repurposed for feature selection. This raises a question: does SHAP-based selection offer advantages over established methods, or does its explanatory elegance come without corresponding gains in predictive utility?

## 1.2 Problem Statement

Despite SHAP's widespread adoption for model interpretation, its effectiveness as a feature selection mechanism remains underexplored in systematic comparisons. Some recent work has questioned whether SHAP importance rankings translate reliably into effective feature subsets [25], while others have reported promising results [17]. Existing literature often showcases SHAP in explanatory contexts without rigorous benchmarking against traditional selectors across diverse datasets and model families. Furthermore, SHAP computation, even with efficient implementations like TreeExplainer [15], incurs non-trivial runtime costs that may not be justified if simpler methods achieve comparable results.

This gap creates uncertainty for practitioners who must choose feature selection strategies under real-world constraints. Without clear evidence on when SHAP-based selection provides meaningful benefits, organizations risk either underutilizing a powerful tool or investing computational resources in approaches that offer little marginal advantage.

## 1.3 Objectives

### 1.3.1 General Objective

This thesis aims to evaluate SHAP-based feature selection for tabular regression problems, comparing its predictive effectiveness, computational cost, and selection stability against established alternatives across multiple datasets and model families.

### 1.3.2 Specific Objectives

1. Conduct a broad benchmark comparing mutual information, recursive feature elimination, tree-based importance, and SHAP-based selection across four datasets and three model types.

2. Quantify the trade-offs between prediction error reduction and execution time for each strategy under varying feature retention levels.

3. Measure selection stability, meaning the consistency of chosen feature subsets across repeated experimental runs, to assess method robustness.

4. Perform a focused evaluation using the best-performing model from the broad benchmark, testing custom SHAP recursive elimination and hybrid strategies.

5. Synthesize practical recommendations for choosing feature selection methods in applied regression projects.

## 1.4 Scope and Limitations

This work is positioned as an applied master's thesis: the goal is not to propose novel algorithms, but to produce a structured, reproducible comparison that yields actionable guidance. The study is limited to supervised tabular regression, four datasets in the broad screening stage, two datasets in the deep-dive stage, and the specific model families and selection strategies implemented in the experimental pipeline.

The analysis does not claim exhaustive coverage of all explainability approaches, all regression estimators, or all hyperparameter optimization schemes. Results should be interpreted as context-sensitive empirical findings applicable to similar tabular regression settings, rather than universal claims. Detailed references are deferred to a subsequent revision pass.

## 1.5 Document Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews the current state of regression modeling, feature selection techniques, and SHAP methodology, establishing the technical foundation for the experiments. Chapter 3 formalizes the research hypotheses and describes the experimental design, including the two-stage approach with broad screening followed by focused evaluation. Chapter 4 presents the methodology in detail and reports the experimental results, with emphasis on performance, stability, and cost trade-offs. Chapter 5 concludes with a synthesis of findings, practical recommendations, and directions for future work.

# Chapter 2

# State of the Art

This chapter reviews the technical foundations underlying this thesis: regression modeling approaches for tabular data, feature selection methodologies, and the SHAP framework with particular attention to TreeExplainer. The review establishes both the theoretical background and practical considerations that inform the experimental design.

## 2.1 Regression Modeling for Tabular Data

Tabular data, structured records with rows representing observations and columns representing features, remains the dominant format in business and scientific applications. Unlike images or text, tabular datasets present heterogeneous feature types, varying scales, and often complex correlation structures that require careful modeling choices.

### 2.1.1 Linear Models

Regularized linear regression, particularly Ridge regression [11], provides a baseline approach with well-understood properties. Ridge regression minimizes the sum of squared residuals plus an L2 penalty on coefficient magnitudes:

$$\hat{\beta} = \arg\min_{\beta} \left( \|y - X\beta\|^2 + \lambda \|\beta\|^2 \right) \tag{2.1}$$

This formulation shrinks coefficients toward zero without eliminating them entirely, providing some implicit feature weighting. Linear models are computationally efficient and interpretable, but may underperform when relationships between features and target are nonlinear.

### 2.1.2 Ensemble Tree Methods

Tree-based ensembles have become the dominant approach for tabular regression [1]. Two architectures are particularly relevant:

**Bagging ensembles**, exemplified by ExtraTrees (Extremely Randomized Trees) [7], construct multiple decision trees on bootstrap samples with additional randomization in split selection. By averaging predictions across trees, these methods reduce variance while maintaining the ability to capture nonlinear relationships and feature interactions.

**Gradient boosting**, exemplified by XGBoost [2], builds trees sequentially where each tree corrects the residual errors of its predecessors. This additive training approach often achieves state-of-the-art performance on structured data, though it introduces more hyperparameters and potential for overfitting compared to bagging methods.

Both ensemble types provide native feature importance measures: for bagging ensembles, importance typically reflects how often a feature is used in splits weighted by the improvement achieved; for gradient boosting, importance can be measured by gain (total reduction in loss from splits using that feature), weight (split frequency), or cover (number of samples affected).

## 2.2 Feature Selection Strategies

Feature selection methods are conventionally categorized into three families based on their relationship with the learning algorithm [8].

### 2.2.1 Filter Methods

Filter methods evaluate features independently of any specific model, typically using statistical measures of relevance. Mutual Information (MI) [3] is a widely used filter criterion that captures arbitrary dependencies between a feature $X$ and target $Y$:

$$MI(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \tag{2.2}$$

For continuous variables, MI is estimated through discretization or kernel density methods. Filter methods are computationally cheap and model-agnostic, but ignore feature interactions and may select redundant variables.

### 2.2.2 Wrapper Methods

Wrapper methods evaluate feature subsets by training the target model and measuring performance. Recursive Feature Elimination (RFE) [9] is a prominent example: starting with all features, RFE iteratively removes the least important feature(s) according to the model's own importance scores, continuing until a desired subset size is reached. While wrapper methods account for model-specific behavior, they are computationally expensive and may overfit to the evaluation metric.

### 2.2.3 Embedded Methods

Embedded methods perform selection as part of model training. L1-regularized regression (Lasso) [24] is the canonical example, where the penalty term drives some coefficients exactly to zero. Tree-based models also embed implicit selection by only splitting on informative features. This thesis treats tree-based native importance as an embedded approach, though it can also serve as input to explicit subset selection.

## 2.3 SHAP: Theory and Computation

SHAP (SHapley Additive exPlanations) [16] provides a unified framework for feature attribution that satisfies desirable theoretical properties from cooperative game theory. Understanding SHAP requires examining both its theoretical foundations and practical computational approaches.

### 2.3.1 Shapley Values in Feature Attribution

The core idea borrows from cooperative game theory, where Shapley values [22] fairly distribute a total payoff among players according to their marginal contributions. In the feature attribution context, features are "players" and the "payoff" is the model prediction minus a baseline (typically the expected prediction).

For a model $f$, the Shapley value $\phi_j$ for feature $j$ is defined as:

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[ f(S \cup \{j\}) - f(S) \right] \tag{2.3}$$

where $N$ is the set of all features, $S$ is a subset excluding feature $j$, and $f(S)$ represents the model prediction when only features in $S$ are "present" (with absent features marginalized out). This formula averages the marginal contribution of feature $j$ over all possible orderings in which features could be added.

Shapley values uniquely satisfy four axioms [5]:

- **Efficiency**: attributions sum to the difference between the prediction and baseline.
- **Symmetry**: features with identical contributions receive identical attributions.
- **Dummy**: features that never change predictions receive zero attribution.
- **Linearity**: attributions combine linearly across additive models.

For feature selection, global importance is typically computed by averaging absolute SHAP values across all observations: $I_j = \frac{1}{n}\sum_{i=1}^{n} |\phi_j^{(i)}|$.

### 2.3.2 TreeExplainer: Efficient SHAP for Tree Ensembles

The naïve computation of Shapley values requires evaluating exponentially many feature subsets, which is intractable for high-dimensional data. TreeExplainer [15] exploits the structure of decision trees to compute exact SHAP values in polynomial time.

The key insight is that a decision tree already partitions the feature space into regions, and the path from root to leaf for any observation defines which features were used. TreeExplainer recursively propagates SHAP values through the tree structure, correctly handling the exponential number of coalitions by clever bookkeeping of how predictions change as features are revealed or hidden.

For an ensemble of $T$ trees, each with $L$ leaves and average depth $D$, TreeExplainer computes exact SHAP values in $O(TLD^2)$ time per observation [15]. Here, $T$ is the number of trees in the ensemble, $L$ is the number of leaves per tree, and $D$ is the tree depth. This is dramatically faster than model-agnostic approaches but still represents non-trivial overhead compared to simply extracting native importance scores.

### 2.3.3 Practical Considerations

Several factors affect SHAP computation in practice:

**Sample size**: computing SHAP for all training observations may be expensive; practitioners often use a representative sample.

**Background data**: SHAP values depend on a reference distribution used to marginalize absent features. TreeExplainer typically uses the training data itself as background.

**Stability**: because tree ensembles are stochastic (due to random splits or bootstrap sampling), SHAP rankings may vary between model fits, potentially affecting selection consistency.

## 2.4 Datasets for Evaluation

Before running long experiments, a pre-study pipeline was executed to download, preprocess, and characterize each dataset. This section summarizes the four datasets selected for the broad benchmark, chosen to span different scales and correlation structures.

Table 2.1: Dataset characteristics after preprocessing.

| **Dataset** | **Rows** | **Num.** | **Cat.** | **Target** | **Domain** |
|---|---:|---:|---:|---:|---|
| Allstate Claims [6] | 188,318 | 14 | 116 | loss | Insurance |
| Bike Sharing [4] | 17,379 | 12 | 0 | cnt | Rentals |
| Communities & Crime [21] | 1,993 | 100 | 0 | ViolentCrimesPerPop | Crime |
| Superconductor [10] | 21,263 | 81 | 0 | critical_temp | Materials |

This collection provides diversity along several axes:

**Scale**: from under 2,000 observations (Communities and Crime) to nearly 200,000 (Allstate), testing whether method behavior differs with sample size.

**Dimensionality**: from 12 features (Bike Sharing) to 130 features (Allstate), covering both low and high-dimensional regimes.

**Correlation structure**: the pre-study correlation heatmaps reveal that some datasets exhibit strong feature clusters (Communities and Crime) while others show more dispersed correlation patterns.

## 2.5 Chapter Summary

This review has established the technical context for the experimental work. Regression on tabular data is best served by ensemble tree methods, though linear baselines remain useful references. Feature selection spans filter, wrapper, and embedded approaches, each with distinct trade-offs. SHAP provides theoretically principled feature attributions, and TreeExplainer makes exact computation tractable for tree models, though not without cost. The selected datasets span a meaningful range of sizes and structures, enabling evaluation of whether method performance generalizes. With this foundation in place, the next chapter formalizes the research hypotheses and experimental design.

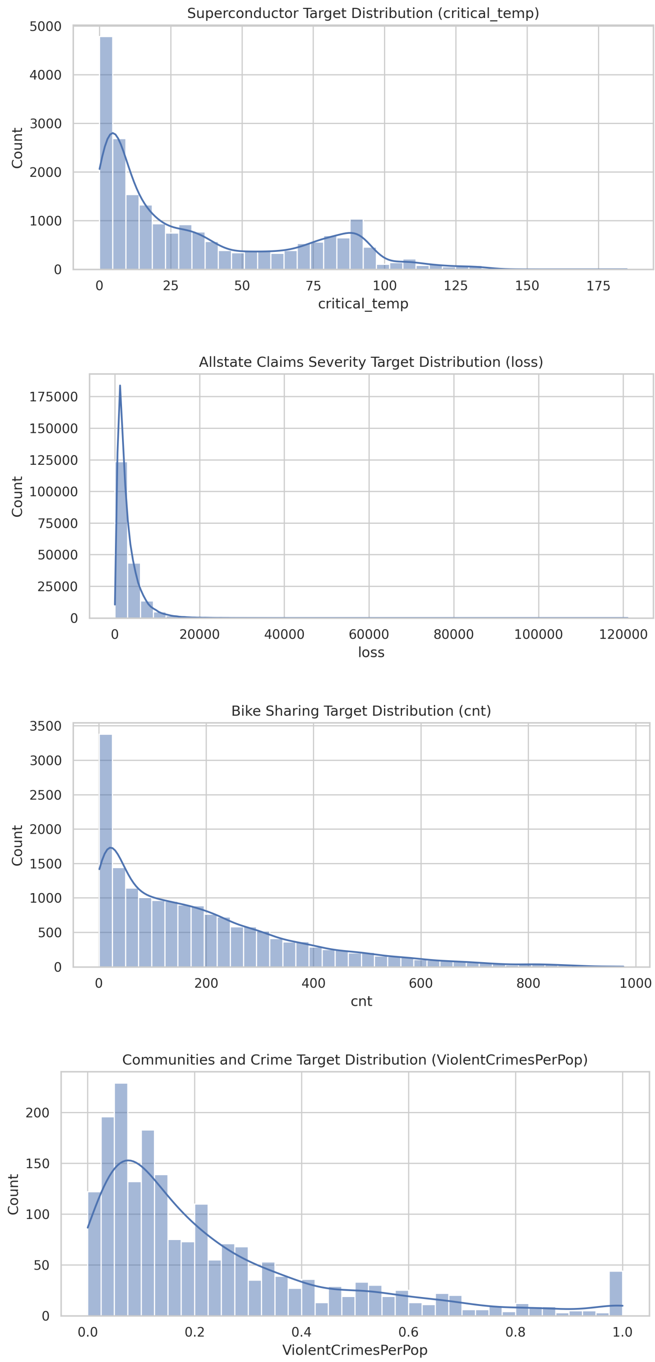


Figure 2.1: Target variable distributions for all four datasets.

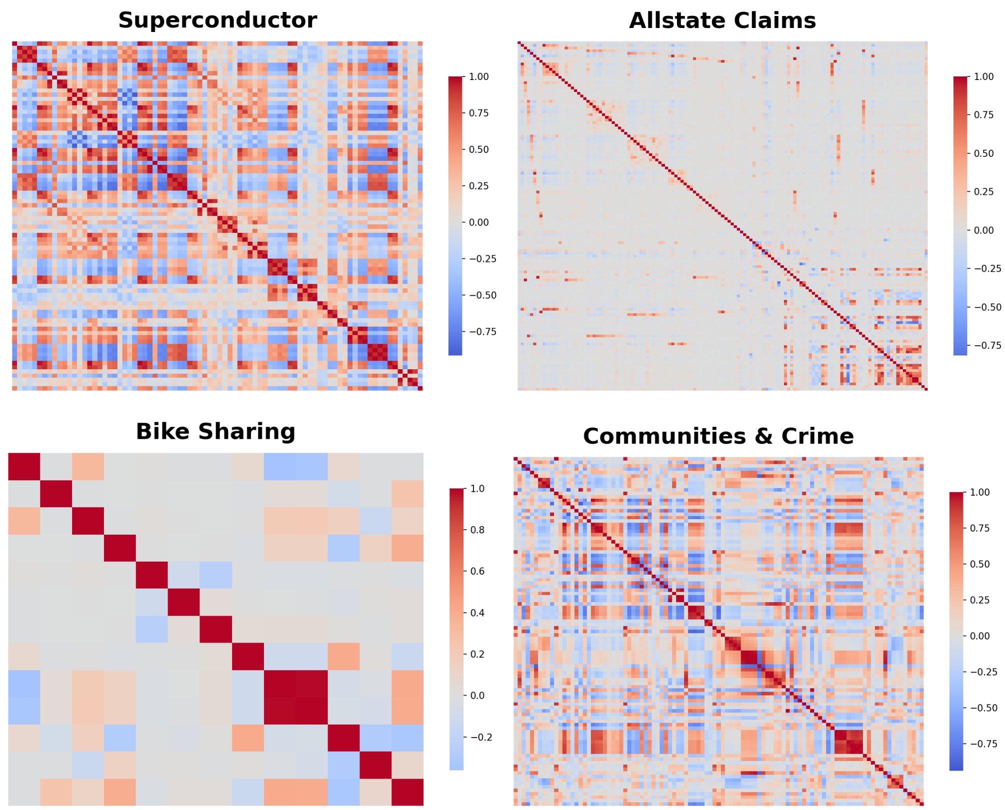


Figure 2.2: Feature correlation heatmaps showing multicollinearity patterns across the four datasets.

# Chapter 3

# Research Hypotheses and Experimental Design

This chapter formalizes the research questions motivating this thesis and describes the experimental protocol designed to address them. The approach follows a two-stage structure: a broad screening benchmark to map general behavior, followed by a focused deep dive to test SHAP-intensive strategies under controlled conditions.

## 3.1 Research Questions

The central inquiry of this thesis can be stated as:

> *Does SHAP-based feature selection provide practically relevant advantages over established alternatives when applied to tabular regression, considering predictive accuracy, selection stability, and computational cost jointly?*

This question decomposes into several testable components:

1. **Effectiveness**: Do SHAP-selected feature subsets yield lower prediction error than subsets chosen by standard methods?

2. **Cost**: What computational overhead does SHAP-based selection incur, and does any performance gain justify this cost?

3. **Stability**: Are SHAP-selected subsets consistent across repeated runs, or does the method introduce additional variability?

4. **Generalization**: Do observed patterns hold across different datasets and model families, or are conclusions dataset-specific?

## 3.2 Hypotheses

Based on the literature review and preliminary analysis, the following hypotheses guide the experimental work:

### 3.2.1 H1: Competitive but Non-Dominant Performance

In the broad benchmark across heterogeneous datasets and model families, SHAP-based feature selection will produce competitive prediction error compared to standard methods (mutual information, recursive feature elimination, tree-based importance), but will not be uniformly dominant. Different methods will excel on different dataset-model combinations, reflecting the no-free-lunch principle [27] in feature selection.

### 3.2.2 H2: Marginal Gains at Intermediate Retention

In focused evaluation with XGBoost, SHAP-based recursive elimination and hybrid strategies may produce small improvements at intermediate feature retention levels (e.g., $k = 0.25$ to $k = 0.50$). At very low retention ($k < 0.15$), aggressive pruning may harm all methods; at full retention ($k = 1.0$), selection becomes irrelevant.

### 3.2.3 H3: Substantial Computational Overhead

SHAP-intensive strategies, particularly those involving iterative SHAP computation (custom SHAP-RFE, hybrid pipelines), will require substantially higher selection time than native importance or simple filter methods.

### 3.2.4 H4: Reduced Stability at Low Retention

Selection stability will decrease as the retention ratio decreases, across all methods. However, SHAP-based methods may exhibit additional instability due to the stochastic nature of tree ensembles and the sensitivity of SHAP rankings to model refitting.

## 3.3 Experimental Design

The experimental protocol consists of two stages designed to balance breadth and depth.

### 3.3.1 Stage 1: Broad Screening Benchmark

The first stage surveys the landscape of feature selection performance across multiple datasets, models, and methods. This screening identifies general patterns and informs the selection of configurations for deeper analysis.

**Datasets**: The four datasets described in Chapter 2 (Allstate Claims, Bike Sharing, Communities and Crime, Superconductor) span a range of sizes and dimensionalities.

**Models**: Three model families are evaluated:

- Ridge regression [11] (linear baseline)
- ExtraTrees [7] (bagging ensemble)
- XGBoost [2] (gradient boosting)

**Feature selection strategies**:

- Mutual Information (MI): filter method based on statistical dependency
- Recursive Feature Elimination (RFE) [9]: wrapper method with iterative removal
- Tree-based Importance (Tree-FI) [1]: embedded method using native importance
- SHAP-based Importance: ranking by mean absolute SHAP value

**Feature retention levels**: Selection is evaluated at $k \in \{0.25, 0.50, 0.75, 1.00\}$, where $k$ represents the proportion of original features retained.

**Evaluation**: Each configuration is run with cross-validation [23] to obtain robust error estimates. Metrics include RMSE (primary), MAE, and MAPE. Selection time and total execution time are recorded. To assess selection stability, each configuration is repeated 5 times with different random seeds, in addition to 10-fold cross-validation within each repetition.

### 3.3.2 Stage 2: Focused Deep Dive

The second stage narrows focus to the model family and datasets that demonstrate the most interesting behavior in Stage 1, based on objective selection criteria: models where feature selection shows actual impact on performance, and datasets with sufficient features to enable fine-grained retention analysis. The specific model and datasets for Stage 2 will be determined based on Stage 1 results.

**Selection strategies**:

- `native_fi`: native feature importance ranking from the selected model
- `custom_shap_rfe`: recursive elimination guided by SHAP scores, removing features iteratively until the target retention is reached
- `hybrid_fi_custom_shap_rfe`: initial pruning using native importance to reduce the feature set, followed by SHAP-guided recursive elimination on the reduced set

- `baseline`: no selection, all features retained ($k = 1.0$)

**Feature retention levels**: A finer grid is used: $k \in \{0.05, 0.10, 0.15, 0.25, 0.50, 1.00\}$.

**Additional metrics**: Beyond prediction error and timing, Stage 2 tracks selection stability via Jaccard similarity between feature subsets across runs, and multicollinearity indicators in the selected subsets.

### 3.3.3 Rationale for Two-Stage Design

A direct exhaustive comparison of all methods across all datasets with fine-grained retention levels would be computationally prohibitive and produce unwieldy results. The two-stage approach serves multiple purposes:

1. **Feasibility**: Stage 1 provides broad coverage with coarse granularity; Stage 2 invests computation where it matters most.

2. **Focus**: By narrowing to a single model family in Stage 2, method effects are isolated from model-family effects.

3. **Relevance**: The hybrid and custom SHAP-RFE strategies are more computationally intensive; testing them on selected configurations ensures practical relevance of the comparison.

## 3.4 Evaluation Metrics

Assessment is structured around three axes, reflecting the thesis objective of evaluating practical utility:

**Predictive effectiveness**: Root Mean Squared Error (RMSE) serves as the primary metric, supplemented by Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). Lower values indicate better performance.

**Selection stability**: Measured by Jaccard similarity [18] between feature subsets selected in different cross-validation folds or repeated runs. Values range from 0 (no overlap) to 1 (identical subsets). Higher stability suggests more robust selection.

**Computational cost**: Measured by selection time (time spent on feature selection) and total execution time (selection plus model training and evaluation). Costs are compared in absolute terms and as ratios between methods.

Comparisons are always interpreted as trade-offs. A method that achieves marginally lower RMSE but requires dramatically higher computation may not be practically preferable.

## 3.5 Implementation Details

All experiments are implemented in Python using scikit-learn [19], XGBoost [2], and the SHAP library [14]. This section describes the specific algorithms used for each selection strategy.

### 3.5.1 Stage 1 Selection Strategies

**Mutual Information (MI)**: Features are ranked by their mutual information with the target variable, computed using scikit-learn's `mutual_info_regression` [8] with default parameters. The top $k$ fraction of features by MI score are retained.

**Recursive Feature Elimination (RFE)**: Using scikit-learn's `RFE` wrapper, features are iteratively removed based on the model's native importance until the target number of features is reached. One feature is removed per iteration.

**Tree-based Importance (Tree-FI)**: For tree-based models, features are ranked by their native importance scores (impurity-based for ExtraTrees, gain-based for XGBoost). The top $k$ fraction is retained. For Ridge regression, coefficient magnitudes serve as importance proxy.

**SHAP Importance**: A model is trained on all features, SHAP values are computed for the training set using TreeExplainer [15] (for tree models) or LinearExplainer (for Ridge), and features are ranked by mean absolute SHAP value. The top $k$ fraction is retained. This is a single-pass approach: SHAP is computed once, features are dropped, and a new model is trained on the reduced set.

### 3.5.2 Stage 2 Selection Strategies

**Native Feature Importance (native_fi)**: Identical to Tree-FI from Stage 1, using XGBoost's gain-based importance.

**Custom SHAP-RFE (custom_shap_rfe)**: A recursive elimination process where at each iteration: (1) the model is trained on current features, (2) SHAP values are computed, (3) the 2 features with lowest mean absolute SHAP are removed, (4) the process repeats until target retention $k$ is reached. This recalculates importance at each step, allowing the model to adapt as features are removed. This approach is conceptually similar to the SHAP-Select method proposed by Kraev et al. [13], which also uses iterative SHAP-based elimination to identify relevant features. Our implementation differs in the elimination step size and stopping criteria, but shares the core principle of recalculating SHAP importance after each removal to capture changing feature interactions.

**Hybrid (hybrid_fi_custom_shap_rfe)**: A two-phase approach that extends the SHAP-RFE concept with a computational optimization. First, native feature importance is used to quickly prune to 50% of features, removing obviously irrelevant variables without SHAP overhead. Then, custom_shap_rfe is applied to the reduced feature set to reach the final target retention. This design tests whether combining fast initial filtering with precise SHAP-based refinement can achieve better cost-effectiveness than pure SHAP-RFE.

### 3.5.3 Computational Cost Implications

The strategies have different computational profiles:

- **MI and Tree-FI**: Fast, single computation of importance scores.
- **Stage 1 SHAP**: Single SHAP computation on full feature set.
- **custom_shap_rfe**: Multiple SHAP computations, one per elimination step. Lower $k$ requires more steps, thus more time.
- **hybrid**: Fewer SHAP computations than custom_shap_rfe since it starts from a pre-pruned set.

### 3.5.4 Reproducibility

Cross-validation uses 10 folds with fixed random seeds. Each configuration is repeated 5 times with different seeds for stability analysis. Results are saved as raw CSV files containing per-fold metrics, along with aggregated summaries. Figures are generated programmatically to ensure consistency.

## 3.6 Chapter Summary

This chapter has established the hypotheses driving the experimental work and described a two-stage protocol designed to evaluate SHAP-based feature selection systematically. The broad screening stage covers multiple datasets and models to identify general patterns, while the focused deep dive tests SHAP-intensive strategies where they are most likely to show value. Evaluation emphasizes practical trade-offs among effectiveness, stability, and cost. The next chapter presents the experimental execution and results.

# Chapter 4

# Experimental Results and Analysis

This chapter reports the outcomes of the two experimental stages and analyzes the findings against the hypotheses established in Chapter 3. The presentation follows the experimental structure: first the broad screening results from Stage 1, then the focused deep-dive results from Stage 2, and finally a synthesis of findings across both stages.

## 4.1 Stage 1: Broad Screening Results

Stage 1 evaluated four feature selection strategies across four datasets and three model families, producing 7,920 individual observations in the summarized outputs. This section presents aggregate findings organized by research question.

### 4.1.1 Predictive Performance

Table 4.1 reports the best-performing strategy for each dataset-model combination, determined by minimum mean RMSE across all retention levels.

Several patterns emerge from this summary:

Table 4.1: Best RMSE by dataset and model in Stage 1. Note: at $k = 1.00$, all strategies retain all features, so "Baseline" indicates no actual selection occurred.

| **Dataset** | **Model** | **Best Strategy** | **k** | **RMSE** |
|---|---|---|---|---|
| Allstate Claims | Ridge | Baseline | 1.00 | 2090.07 |
| Allstate Claims | ExtraTrees | Baseline | 1.00 | 1992.76 |
| Allstate Claims | XGBoost | SHAP | 0.50 | 1897.11[1] |
| Bike Sharing | Ridge | Baseline | 1.00 | 141.04 |
| Bike Sharing | ExtraTrees | Baseline | 1.00 | 40.83[2] |
| Bike Sharing | XGBoost | Baseline | 1.00 | 42.57 |
| Communities & Crime | Ridge | Baseline | 1.00 | 0.136 |
| Communities & Crime | ExtraTrees | Tree-FI | 0.50 | 0.134 |
| Communities & Crime | XGBoost | SHAP | 0.75 | 0.138 |
| Superconductor | Ridge | Baseline | 1.00 | 17.65 |
| Superconductor | ExtraTrees | MI | 0.75 | 9.11 |
| Superconductor | XGBoost | Tree-FI | 0.75 | 9.98 |

**Baseline often wins**: In 7 of 12 configurations, the best RMSE is achieved at $k = 1.00$, meaning no feature selection occurred. When all features are retained, all strategies produce identical results since no selection is performed. This is labeled "Baseline" in the table to clarify that these are not feature selection victories.

**Ridge never benefits from selection**: The linear model achieves its best RMSE at $k = 1.00$ across all four datasets. Ridge regularization [11] already handles multicollinearity and noisy features effectively, making explicit feature removal unnecessary or harmful.

**Tree-based models sometimes benefit**: In contrast, ExtraTrees and XGBoost achieve their best performance at lower retention levels in some cases. XGBoost on Allstate prefers $k = 0.50$ with SHAP selection, and ExtraTrees on Communities prefers $k = 0.50$ with Tree-FI. This suggests tree-based models can benefit from removing noisy or redundant features.

**When selection helps, different methods win**: Among configurations where $k < 1.00$ is optimal, SHAP wins twice (Allstate and Communities with XGBoost), Tree-FI wins twice (Communities with ExtraTrees, Superconductor with XGBoost), and MI wins once (Superconductor with ExtraTrees). No single selection method dominates, consistent with the no-free-lunch principle [27].

Figure 4.1 visualizes RMSE by strategy across datasets, illustrating the variability in method rankings.

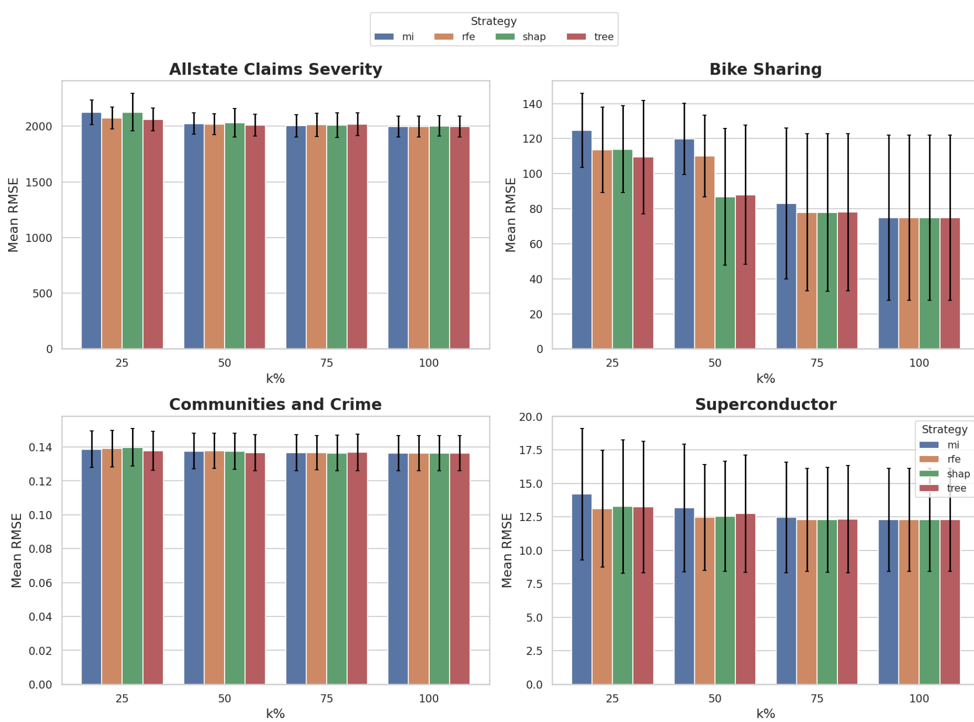


Figure 4.1: Stage 1 RMSE comparison by strategy across all four datasets.

### 4.1.2 Computational Cost

Figure 4.2 shows the distribution of computational costs across configurations. The cost patterns reveal important nuance: SHAP selection time varies dramatically by model. For Ridge, SHAP computation is near-instantaneous (linear SHAP is analytically tractable [16]). For ExtraTrees, SHAP incurs substantial overhead, roughly 6-7x the cost of MI. For XGBoost, the TreeExplainer efficiency keeps SHAP competitive with other methods.

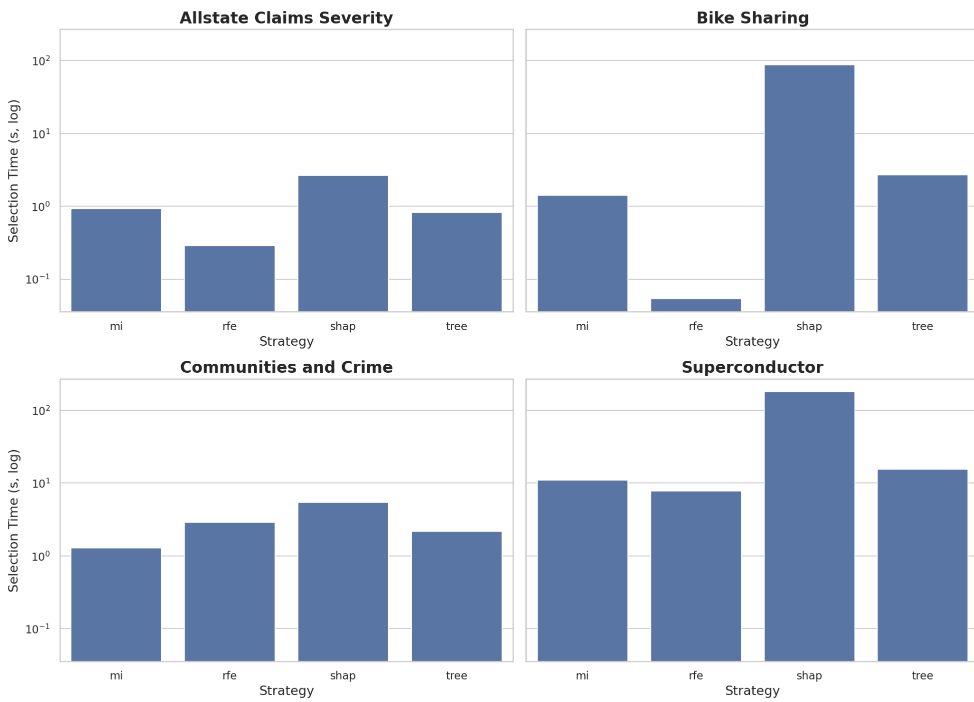


Figure 4.2: Stage 1 computational cost by strategy and dataset.

### 4.1.3 Selection Stability

Table 4.2 summarizes Jaccard stability [18] scores by strategy and retention level, averaged across datasets.

Table 4.2: Mean Jaccard stability by strategy and retention level in Stage 1.

| **Strategy** | **k=0.25** | **k=0.50** | **k=0.75** | **k=1.00** |
|---|---|---|---|---|
| MI | 0.81 | 0.83 | 0.96 | 1.00 |
| RFE | 0.73 | 0.79 | 1.00 | 1.00 |
| SHAP | 0.63 | 0.77 | 0.97 | 1.00 |
| Tree-FI | 0.75 | 0.87 | 0.95 | 1.00 |

Stability increases with retention level across all methods, as expected, since larger subsets have more overlap by construction. At $k = 0.25$, SHAP shows the lowest stability (0.63), supporting hypothesis H4 that SHAP-based selection may introduce additional variability. By $k = 0.75$, all methods converge to high stability.

### 4.1.4 Stage 1 Summary

The broad screening confirms that SHAP-based selection can be competitive but is not systematically superior. XGBoost emerges as the most consistent performer among models, making it the natural choice for the focused Stage 2 analysis. The computational overhead of SHAP depends heavily on the model family, with TreeExplainer providing efficiency for gradient boosting.

## 4.2 Stage 2: Focused Deep-Dive Results

Stage 2 applies three feature selection strategies plus a baseline to XGBoost on two selected datasets: Superconductor and Allstate Claims. A finer retention grid ($k \in \{0.05, 0.10, 0.15, 0.25, 0.50, 1.00\}$) enables more detailed analysis.

**Dataset selection rationale**: Superconductor and Allstate Claims were chosen for Stage 2 because they exhibited meaningful performance variation across retention levels in Stage 1, unlike Bike Sharing where all strategies performed similarly. Additionally, both datasets have sufficient dimensionality (81 and 130 features respectively) to support finer-grained retention values without reducing to trivially small feature sets.

### 4.2.1 Implementation Note: SHAP Selection Approaches

The SHAP-based selection differs between stages. In Stage 1, SHAP importance is computed once on the full model, features are ranked, and the bottom $(1 - k)$ fraction is dropped in a single pass before retraining. This approach is computationally efficient but potentially unstable since the importance ranking is based solely on the full-feature model.

In Stage 2, the custom_shap_rfe strategy uses a recursive approach: SHAP values are computed, the bottom 2 features are dropped, the model is retrained, and the process repeats until the target retention level $k$ is reached. This iterative elimination recalculates feature importance at each step, allowing the model to adapt as features are removed. While slower, this approach tends to produce more stable feature rankings since each elimination decision is made with updated importance scores.

### 4.2.2 Superconductor Dataset

Table 4.3 presents RMSE for each strategy and retention level on the Superconductor dataset.

Table 4.3: Superconductor results: RMSE by strategy and retention level.

| **Strategy** | **k** | **RMSE** |
|---|---|---|
| native_fi | 0.05 | 13.44 |
| custom_shap_rfe | 0.05 | 13.48 |
| hybrid | 0.05 | 13.51 |
| native_fi | 0.10 | 11.20 |
| custom_shap_rfe | 0.10 | 11.10 |
| hybrid | 0.10 | 11.15 |
| native_fi | 0.25 | 10.30 |
| custom_shap_rfe | 0.25 | 10.35 |
| hybrid | 0.25 | 10.34 |
| native_fi | 0.50 | 10.08 |
| custom_shap_rfe | 0.50 | 10.06 |
| hybrid | 0.50 | 10.05 |
| baseline | 1.00 | 9.94 |

Key observations:

**Baseline wins at full retention**: The best absolute RMSE (9.94) is achieved with all features retained, suggesting that for this dataset, feature selection does not improve over the full model.

**Marginal differences between selection strategies**: At each retention level, native_fi, custom_shap_rfe, and hybrid produce nearly identical RMSE (differences $< 0.1$). The sophisticated SHAP-based approaches do not outperform simple native importance.

Figure 4.3 visualizes RMSE by retention level.

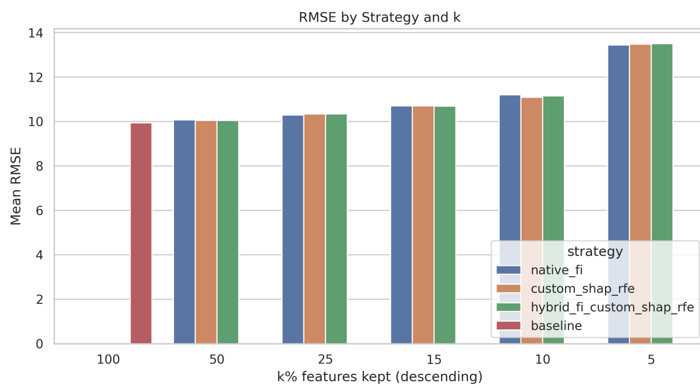


Figure 4.3: Superconductor deep-dive: RMSE by retention level for each strategy.

### 4.2.3 Allstate Claims Dataset

Table 4.4: Allstate Claims results: RMSE by strategy and retention level.

| **Strategy** | **k** | **RMSE** |
|---|---|---|
| native_fi | 0.05 | 2198.79 |
| custom_shap_rfe | 0.05 | 2277.81 |
| hybrid | 0.05 | 2224.63 |
| native_fi | 0.10 | 2062.39 |
| custom_shap_rfe | 0.10 | 2088.88 |
| hybrid | 0.10 | 2016.12 |
| native_fi | 0.25 | 1929.38 |
| custom_shap_rfe | 0.25 | 1925.70 |
| hybrid | 0.25 | 1932.26 |
| native_fi | 0.50 | 1912.92 |
| custom_shap_rfe | 0.50 | 1913.03 |
| hybrid | 0.50 | 1913.72 |
| baseline | 1.00 | 1913.12 |

Table 4.4 presents corresponding results for the Allstate dataset. On Allstate, patterns are similar to Superconductor:

**Feature selection matches baseline**: At $k = 0.50$, native_fi achieves RMSE

1912.92, essentially identical to the baseline (1913.12). Selection provides no meaningful improvement.

**SHAP-RFE occasionally wins**: At $k = 0.25$, custom_shap_rfe achieves the best RMSE (1925.70), marginally better than native_fi (1929.38). This is consistent with hypothesis H2, showing small gains at intermediate retention.

**Hybrid shows no advantage**: At $k = 0.50$, hybrid produces the worst RMSE (1913.72) among selection methods, offering no benefit over simpler approaches.

Figure 4.4 visualizes RMSE by retention level.

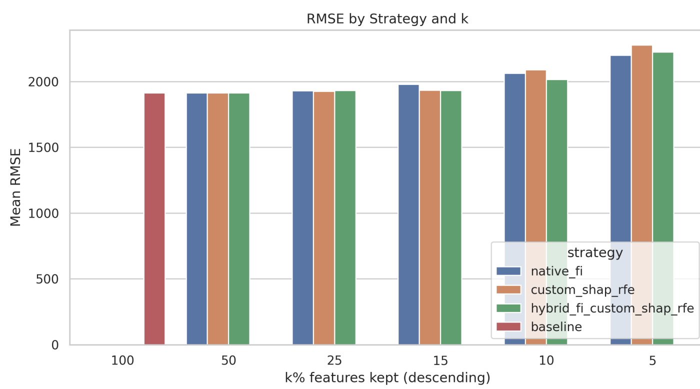


Figure 4.4: Allstate Claims deep-dive: RMSE by retention level for each strategy.

### 4.2.4 Computational Cost in Stage 2

The computational cost depends on the number of SHAP computation iterations required. Table 4.5 shows the expected iterations for each strategy on the Superconductor dataset (81 features), removing 2 features per elimination step.

**native_fi** requires zero SHAP iterations, using only native XGBoost impor-

Table 4.5: SHAP iterations by strategy and retention level (Superconductor, 81 features).

| Strategy | k=0.05 | k=0.10 | k=0.25 | k=0.50 | k=1.00 |
|---|---|---|---|---|---|
| native_fi | 0 | 0 | 0 | 0 | 0 |
| hybrid | 19 | 18 | 15 | 10 | 0 |
| custom_shap_rfe | 38 | 36 | 30 | 20 | 0 |

tance. Cost is constant regardless of $k$.

**custom_shap_rfe** starts from all 81 features and eliminates 2 per iteration until reaching target $k$. For $k = 0.05$ (4 features), this requires $(81 - 4)/2 \approx 38$ iterations. Each iteration involves model training and SHAP computation.

**hybrid** first prunes to 50% using native importance (instant), then applies SHAP-RFE on the reduced set. For $k = 0.05$, it starts from 40 features: $(40 - 4)/2 = 18$ iterations.

## 4.3 Synthesis and Hypothesis Evaluation

The combined evidence from both stages supports the following conclusions:

**H1 (Competitive but non-dominant)**: *Supported.* SHAP-based selection wins in 2 of 12 Stage 1 configurations, both with XGBoost. It is competitive but not dominant.

**H2 (Marginal gains at intermediate retention)**: *Partially supported.* Small RMSE improvements appear at $k = 0.25$ to $k = 0.50$ in select cases (Allstate XGBoost with SHAP in Stage 1; custom_shap_rfe at $k = 0.25$ in Stage 2 Allstate). However, these gains are minimal ($< 1\%$ relative improvement).

**H3 (Substantial computational overhead)**: *Supported*. The custom_shap_rfe strategy requires multiple SHAP computations, making it substantially slower than native_fi, especially at low retention levels. The hybrid strategy reduces this overhead by pre-pruning with native importance before SHAP iterations, but still incurs more cost than native_fi alone.

**H4 (Reduced stability at low retention)**: *Partially supported*. Stability does decrease at low $k$ for all methods, but SHAP-based methods do not show dramatically worse stability than alternatives. In some cases (custom_shap_rfe on Allstate), SHAP stability is high.

## 4.4 Threats to Validity

Several limitations affect the generalizability of these findings:

**Dataset selection**: Four datasets cannot represent all tabular regression problems. Different domains or data characteristics might yield different conclusions.

**Hyperparameter choices**: Models were run with default hyperparameters throughout both stages. Joint optimization of feature selection and model hyperparameters might change relative rankings.

**Implementation details**: The custom SHAP-RFE and hybrid strategies are specific implementations. Alternative designs (e.g., different step sizes, pruning thresholds) might perform differently.

## 4.5 Chapter Summary

The experimental results demonstrate that SHAP-based feature selection, while theoretically principled and occasionally achieving the best RMSE, does not provide consistent practical advantages over simpler alternatives. Mutual Information and native tree importance frequently match or exceed SHAP performance at lower computational cost. The hybrid strategy, designed to combine the best of both worlds, instead combines computational expense with marginal performance. These findings suggest that practitioners should default to simpler methods and reserve SHAP-based selection for cases where its value has been demonstrated on the specific task at hand.

# Chapter 5

# Conclusions and Future Work

This thesis evaluated SHAP-based feature selection for tabular regression through a two-stage protocol. Stage 1 screened four selection strategies across four datasets and three model families. Stage 2 performed focused analysis of SHAP-intensive methods using XGBoost on two datasets.

The central finding is that SHAP-based feature selection does not provide consistent advantages over simpler alternatives in the settings studied:

**Predictive effectiveness**: SHAP achieved the best RMSE in only 2 of 12 dataset-model configurations in Stage 1, both with XGBoost. In Stage 2, custom SHAP recursive elimination occasionally outperformed native importance at intermediate retention levels, but gains were marginal, typically less than 1%. In many configurations, the full-feature baseline matched or exceeded all selection methods.

**Computational cost**: SHAP-based strategies involve additional computation compared to native importance, particularly for bagging ensembles where TreeExplainer is less efficient. The hybrid strategy adds complexity without corresponding accuracy gains.

**Selection stability**: SHAP-based methods showed somewhat lower stability at aggressive retention levels ($k = 0.25$) in Stage 1, though this pattern was not dramatic. Stability concerns do not strongly differentiate SHAP from alternatives.

## 5.1 Interpretation

These results do not suggest SHAP is without value; it remains powerful for model explanation [15]. However, the evidence does not support treating SHAP importance as a superior selection criterion for general regression practice, consistent with recent findings [25].

The lack of consistent SHAP advantage likely reflects:

- **Redundancy with native importance**: For tree-based models, SHAP values and native importance often correlate strongly [5], so additional SHAP computation may not surface different rankings.
- **Model robustness**: Modern ensemble methods [1] are robust to irrelevant features, reducing the benefit of explicit selection.
- **Dataset characteristics**: The datasets studied may not exhibit conditions where fine-grained SHAP rankings matter.

## 5.2 Practical Recommendations

Based on the experimental evidence:

1. **Start simple**: Begin with a full-feature baseline. Many datasets do not benefit from explicit selection.

2. **Use low-cost methods first**: Mutual information or native importance rankings provide competitive results at minimal cost.

3. **Reserve SHAP for targeted use**: SHAP-based selection may be justified when there is specific evidence it helps, or when interpretability is the primary goal.

4. **Avoid expensive hybrids without validation**: Complex strategies do not automatically outperform simpler methods. Justify their cost with demonstrated gains.

5. **Consider the full trade-off**: A method achieving marginally lower RMSE but requiring significantly more time may not be preferable in practice.

## 5.3 Limitations

- **Scope**: Four datasets and three model families cannot represent all regression problems. Results may differ on time-series data or high-dimensional settings.

- **Hyperparameters**: Models used default or lightly-tuned configurations. Joint optimization might reveal different patterns.

- **Strategy implementations**: The custom SHAP-RFE and hybrid approaches are specific implementations; alternatives might perform differently.

- **Metrics**: The study focused on RMSE, stability, and runtime. Interpretability or regulatory requirements might favor different choices.

## 5.4 Future Work

**Broader coverage**: Evaluating on additional domains (healthcare, finance, industrial sensors) would test generalizability.

**Alternative architectures**: Extending to other boosting implementations (LightGBM [12], CatBoost [20]) would assess transferability.

**Joint optimization**: Nested hyperparameter tuning where selection parameters are optimized alongside model parameters could reveal missed interactions.

**Cost-aware selection**: Adaptive strategies that trigger SHAP computation only when ranking uncertainty is high might reduce overhead while preserving benefits.

In closing, this thesis asked whether SHAP-based feature selection justifies its computational cost in tabular regression. The evidence suggests SHAP provides at best marginal advantages over simpler alternatives. For practitioners under computational constraints, simpler methods remain the pragmatic default. SHAP's value lies primarily in its explanatory power, not as a selection criterion.

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